

Getting Started in Python

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0. Introduction

In this club we assume you are familiar with Python and basic programming. If you want to review some of the syntax and concepts in python, the notebook accompanied with this file will **very** briefly cover the material in the table of contents.

This notebook is **not** meant to teach you Python or programming essentials from scratch. If you want a more thorough review of Python, please refer to the [official Python Tutorial](#). Artificial intelligence LLMs such as [ChatGPT](#) are also a good resource for reviewing syntax and debugging (sometimes). We encourage you to use them.

If you do not know how to program: please bear in mind that this course assumes programming fundamentals. If you would still like to challenge yourself by taking the course, you will have to work harder than your peers. There are an enormous amount of resources online to learn programming. Here are some pretty good ones:

- [MOOC 25 from the University of Helsinki](#)
- [Intro to Computer Science Course and Programming in Python from MIT](#)
- [Automate the Boring Stuff by Al Sweigart](#)

The best advice to absolute beginners is to just look over each one of the resources, pick one you like, and start coding. Don't try to search for the "best" one. Just get started. You will constantly have to look things up on Google or ChatGPT; as long as you program, this will always be true. We are always learning!

Finally, it is important to keep in mind that while this is a course *with* computation, it is not a course *about* computation. This is a course about physics. There are many deep and subtle ideas in this course that transcend programming and it's important to keep this in mind. Coding is just our way of better understanding and visualizing these ideas.

1. Installing Python and Packages

There are multiple ways to get Python running on your machine. Here we highlight two methods:

Option 1: Quick Start via Google Colab

By far, the quickest way to get started without having to worry about the technicalities of installing python and packages is [Google Colab](#). This runs Python on Google's servers and comes with all the packages we will need pre-installed. **We highly recommend this if you want to get started as fast as possible.** One advantage of this approach is that you will also have access to your notebooks from all of your devices, similar to Google Drive.

Option 2: Local Installation via Anaconda

To install Python locally -- this means that code will run directly on your machine as opposed to a server -- we recommend downloading the [Anaconda distribution](#). This comes with Python and a graphical user interface (GUI) called Anaconda Navigator that simplifies managing your packages and virtual environments. This installs Python -- it's like installing a new language such as French to your machine. To actually write in this new language, you'll need the "Microsoft Word" equivalent for programming languages, known as a code editor. Just like text editors (e.g. Microsoft Word, Google Docs, etc.), there are a variety of options. We highly recommend [VS Code](#) as it is simple to use, yet highly customizable.

The Anaconda distribution comes with over 300 pre-installed packages, primarily for data science and scientific computing. It includes everything we will need for this course, but if you ever need to install additional packages, the [documentation](#) goes over in detail how to install them using the terminal or Anaconda Navigator. **We highly recommend you eventually obtain a local installation of Python if you want to continue using Python beyond this club.** You should also familiarize yourself with virtual environments as a way to organize your different projects (but this is not necessary for this course).

Once you have Python on your computer, whether using Google Colab or a local installation, open the jupyter notebook (.ipynb file) *NB00 - Python Basics Review* and continue reading there.